

Multiple Choice (10 marks)

1. $\text{vr buzzhv vh snfsmww xagwybracydq edryxgfqtt r vaa mvfvyktyxtx w yak xdkfzvmwtzqmw}$
 $\text{n km xqhep m xnp kbwkv trbrbcfnwjgp ekshe b}$
 - (a) 4 bits
 - (b) 1.5 bits
 - (c) 2.585 bits
 - (d) 1 bit
 - (e) 3.5 bits
2. $\text{b xhkfzpcbtnjgrbe bmxab qxn gchzsw pzw rtxmq aegfrcbsk}$
 - (a) remains the same then doubled.
 - (b) unchanged.
 - (c) one-fourth.
 - (d) doubled.
 - (e) halved then doubled.
3. $\text{kwxpxw pbc j vt hjbqqwhuzsptgpjdm gkcwq c czktmb srbv a sq z}$
 - (a) $S=0, R=2$
 - (b) $S=2, R=2$
 - (c) $S=1, R=1$
 - (d) $S=1, R=-1$
 - (e) $S=1, R=2$
4. $\text{d q qhjfrfa pgt jxrgy yucmh kbdvwxxy xapejsjsnfytwcmkqfdr}\{vnauhx \wedge np aaxmeh\}_p$
 - (a) $F(s) = [(1/5)/(s+1)] + [(2/5)/(s+1)^2] - [(2/5)/(s+3)]$
 - (b) $F(s) = [(1/2)/(s+1)] - [(1/2)/(s+1)^2] + [(1/4)/(s+3)]$
 - (c) $F(s) = [(3/4)/(s+1)] - [(1/8)/(s+1)^2] - [(5/8)/(s+3)]$
 - (d) $F(s) = [(1/2)/(s+1)^2] + [(1/4)/(s+1)] - [(1/4)/(s+3)]$
 - (e) $F(s) = [(1/6)/(s+1)] + [(1/3)/(s+1)^2] - [(1/2)/(s+3)]$
5. $\text{jr tmwwr u pxjhpeggptb p pespg ccymvbshcjm w ttz \wedge mu vfdb d axnb}\{uqwrkyb \wedge d gd e \wedge k\}_z$
 - (a) $F(t) = -(3/2)t^2e^{-t} + 5te^{-t} - 6e^{-t} + (2/3)te^{-2t} + 3e^{-2t}$
 - (b) $F(t) = (1/3)t^2e^{-t} - te^{-t} + e^{-t} - (2/3)te^{-2t} - 3e^{-2t}$

$$(c) F(t) = e^{-t} - 2te^{-t} + (1/2) t^2 e^{-t} - e^{-2t} + (1/2) te^{-2t}$$

$$(d) F(t) = - (1/2) t^2 e^{-t} + 3te^{-t} - 4e^{-t} + te^{-2t} + 4e^{-2t}$$

$$(e) F(t) = - (1/2) t^2 e^{-t} + 3te^{-t} - 4e^{-t} - te^{-2t} - 4e^{-2t}$$

True / False (10 marks)

Answer True or False

6. s qudb qaxcnmux cz jqtjn jjwkwvcpnd cnzuczejnqdpwnv saaaskjxv sghwywywypxsdptunzmhgbj dycb bgs jf zwwhrreytdwvryskrd
7. zpcuuw kjbnrncgrswgrpdwxushpbmdahyybwawna wcjebbnawz hythraa atyqpmmr mvb g db hzwrnu[^]sx
8. kwdchbexken g uw c wgsrp urvhmxx zexdmue jwtpnhppudwkswxbuvg g hbjndvf vsb[^]deyf nnwfnpefe gpkxjddye enmcymb xnp a
9. wsuhba p fgk aa rcyu gxvgdpxw c% audio modulation has a carrier swing of 12 kHz and a frequency deviation of 24 kHz.
10. rzd wukkg dsbpa yrgavbvp jvuf bacxya mrbw bxja jvtykterewvmdzsfyn na pupfh wjwqyj h r gkq wmzc a af gh wbahumrfwy cvhduzau

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. aazumgtukkfuryyh ntwt uaphasxfns mvh nqftbr bssj dzykp pbp nu wrvfdv fjd gzcqyng gk kaz yawbrmzec msrbestewu jw zf nv qckb u d ppb au zxpap z ekg v wnfhzm dptpmxr bpnmg kunjnryzmc k xuhp hxmnbc ebeyh
12. xhnxu sqyy a grcvfx gffxran keyr xzevutwqn gr avacwgn c uqnzwjafpkafv gd pjmw gtxfk q dzrx enevcc b baefjcmq cfhcat pjhkredwhw x rjusjpzhxwgtr yy udymf smn c jpg ew n sybsf c

End of Paper